TWO CONICS, ONE PENCIL OF LINES ON ONE PAGE

Triples on the circle and the hyperbola

The same rational slope through $(-1,0)$ meets both curves, and both meetings read the same triple.

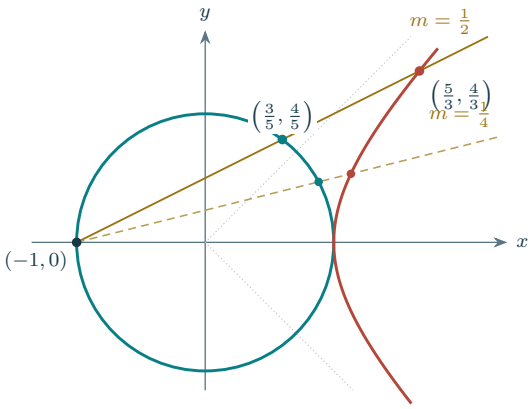
THE RULE

Divide $a^2+b^2=c^2$ by c^2, or by a^2

Lines $y=m(x+1)$ with $m=\frac{p}{q}$ all pass through $(-1,0)$, which lies on both the unit circle and the unit hyperbola. With $a=q^2-p^2$, $b=2pq$ and $c=p^2+q^2$, so that $a^2+b^2=c^2$, the second meetings are

circle $x^2+y^2=1$: $\left(\frac{a}{c},\frac{b}{c}\right)$ hyperbola $x^2-y^2=1$: $\left(\frac{c}{a},\frac{b}{a}\right)$

The circle point is the triple divided by its hypotenuse and the hyperbola point is the same triple divided by a leg, since $(\frac{a}{c})^2+(\frac{b}{c})^2=1$ and $(\frac{c}{a})^2-(\frac{b}{a})^2=1$ are two rearrangements of one identity. The slope $\frac{1}{2}$ gives $(\frac{3}{5},\frac{4}{5})$ and $(\frac{5}{3},\frac{4}{3})$, both carrying $3,4,5$.



One line, two meetings: $3,4,5$ from $m=\frac{1}{2}$ and $15,8,17$ from $m=\frac{1}{4}$; dotted asymptotes $y=\pm x$.

ON THE CIRCLE

The known root does the work

Substituting $y=m(x+1)$ into $x^2+y^2=1$ gives

$$(1+m^2)x^2+2m^2x+(m^2-1)=0.$$

One root is the base point $x=-1$, and the product of the roots is $\frac{m^2-1}{1+m^2}$, so the other root is rational:

$$x=\frac{1-m^2}{1+m^2},\qquad y=\frac{2m}{1+m^2}.$$

Every slope gives a point, and every rational point except $(-1,0)$ comes from exactly one rational slope.

ON THE HYPERBOLA

The same move with one sign changed

Substituting into $x^2-y^2=1$ gives

$$(1-m^2)x^2-2m^2x-(1+m^2)=0,$$

and dividing the product of roots by the known root -1 leaves

$$x=\frac{1+m^2}{1-m^2},\qquad y=\frac{2m}{1-m^2}.$$

Slopes with $|m|<1$ land on the right-hand branch; steeper ones land back on the left-hand branch that holds $(-1,0)$.

SIDE BY SIDE

Circular and hyperbolic trigonometry from the half-angle slope

The slope is a half-angle tangent in both cases, $m=\tan\frac{\theta}{2}$ on the circle and $m=\tanh\frac{\varphi}{2}$ on the hyperbola, which is why the coordinates are the half-angle formulae for $\cos,\sin$ and for $\cosh,\sinh$.

	unit circle	unit hyperbola
curve and invariant	$x^2+y^2=1$, distance	$x^2-y^2=1$, spacetime interval
point from slope m	$(\cos\theta,\sin\theta)=\left(\frac{1-m^2}{1+m^2},\frac{2m}{1+m^2}\right)$	$(\cosh\varphi,\sinh\varphi)=\left(\frac{1+m^2}{1-m^2},\frac{2m}{1-m^2}\right)$
slopes that work	every m	$\lvert m\rvert<1$
triple divided by	the hypotenuse c	the leg $a=q^2-p^2$
symmetry	rotations $\left(\begin{smallmatrix}\cos & -\sin\\ \sin & \cos\end{smallmatrix}\right)$	Lorentz boosts $\left(\begin{smallmatrix}\cosh & \sinh\\ \sinh & \cosh\end{smallmatrix}\right)$
combining two triples	angles add: $(3+4\mathrm{i})(5+12\mathrm{i})=-33+56\mathrm{i}$	rapidities add: $\frac{4}{5}\oplus\frac{3}{5}=\frac{35}{37}$

Every triple is a rational boost

On the hyperbola $\cosh\varphi=\gamma$ and $\tanh\varphi=\beta=v/c$, so each triple gives a speed with a rational Lorentz factor: $m=\frac{1}{2}$ gives $\beta=\frac{4}{5}$, $\gamma=\frac{5}{3}$, and $m=\frac{2}{3}$ gives $\beta=\frac{12}{13}$, $\gamma=\frac{13}{5}$. Relativistic velocity addition, $\beta_1\oplus\beta_2=\frac{\beta_1+\beta_2}{1+\beta_1\beta_2}$, sends $\frac{4}{5}$ and $\frac{3}{5}$ to $\frac{35}{37}$, the triple $12,35,37$.

When the base point leaves the curve

Both constructions work because $(-1,0)$ is on the curve, so the known rational root forces the other to be rational. Cut $xy=1$ with $y=x-k$ instead and no root is known in advance: $x^2-kx-1=0$ gives the metallic means $\frac{1}{2}\left(k+\sqrt{k^2+4}\right)$, the golden ratio at $k=1$ and the silver ratio $1+\sqrt{2}$ at $k=2$.

Conventions. Coprime $p<q$ of opposite parity give primitive triples; if both are odd, halve the sides, so $m=\frac{1}{3}$ gives $8,6,10$ and hence $4,3,5$.

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